

Data Cleaning And Filling

Task 1

Create a DataFrame:

```
data = {
    "Name": ["Raj", "Simran", None, "Ankit"],
    "Age": [22, None, 25, None],
    "Marks": [80, 90, None, 85]
}
```

☞ **Do the following:**

1. Fill missing `Name` with `"Unknown"`
 2. Fill missing `Age` with **mean**
 3. Fill missing `Marks` with **median**
-

Task 2

Given:

```
df = pd.DataFrame({
    "Product": ["Pen", "Pencil", "Eraser", None],
    "Price": [10, None, 5, None]
})
```

☞ **Tasks:**

- Fill missing `Product` with `"Not Available"`
 - Fill missing `Price` with `0`
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☐ Level 2: Column-wise Rules (`dictionary fillna`)

Task 3

Create a DataFrame:

```
data = {
    "Name": ["A", None, "C", None],
    "Age": [20, None, 22, None],
    "Salary": [30000, None, None, 40000]
}
```

🔗 Task:

Fill missing values using **one fillna() call**:

- Name → "No Name"
 - Age → **column mean**
 - Salary → 25000
-

Task 4

Given:

```
df = pd.DataFrame({
    "City": ["Delhi", None, "Mumbai", None],
    "Population": [None, 15000000, None, 12000000]
})
```

🔗 Tasks:

- Replace missing City with "Unknown City"
 - Replace missing Population with **median**
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📦 Level 3: Conditional Fill using loc

Task 5

```
df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Age": [None, 25, None, None],
    "City": ["Delhi", "Mumbai", "Delhi", "Mumbai"]
})
```

🔗 Tasks:

- If City == 'Delhi', fill missing Age with 18
- If City == 'Mumbai', fill missing Age with 30

Task 6

```
df = pd.DataFrame({
    "Employee": ["E1", "E2", "E3", "E4"],
    "Salary": [None, 40000, None, None],
    "Department": ["HR", "IT", "HR", "IT"]
})
```

☞ Tasks:

- Fill missing Salary:
 - HR → 25000
 - IT → 35000